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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PIPING & INSTRUMENTATION DIAGRAM: DECOMPOSER REACTOR D-2304 & CIRCULATION SYSTEM

Drawing Number: PID-23-0013 | Revision: Z1 | Process Unit: Unit 2300

1. EQUIPMENT TITLE BLOCK & DESIGN RATING CALLOUTS ON DRAWING PID-23-0013

- Equipment Tag: D-2304 (Decomposer Reactor)

* P&ID Title Block Mechanical Rating Note: Internal Design Pressure = 12.2 kg/cm2g at 250 degC

* ENGINEERING AUDIT NOTE (CONFLICT): P&ID PID-23-0013 specifies D-2304 Internal Design Pressure as 12.2 kg/cm2g

- Equipment Tag: E-2307 (Decomposer Reactor Circulation Cooler)

* Duty: 4.85 Gcal/hr | Tube Design: 11.0 kg/cm2g at 250 degC | Shell Design: 7.0 kg/cm2g at 100 degC

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PID-23-0013 PAGE 2 OF 3 - PIPING TOPOLOGY & PROCESS CONNECTIONS

- Line 4"-P-2302-316L (Stream S229): From Preflash Column V-2301 & Pump P-2301A/B to D-2304 Nozzle N1
- Line 1"-P-2311-316L (Stream S310): Sulfuric Acid Catalyst Feed through SIL-2 Shutdown Valve XV-1305 to D-2304
- Line 10"-P-2307-316L (Stream S379): From D-2304 Nozzle N2 via Circulation Pump P-2304A/B and Flow Meter FT-1304
- Line 10"-P-2308-316L (Stream S380): Cooled Recirculation Return from E-2307 Tube Outlet to D-2304 Nozzle N3
- Line 4"-P-2306-316L (Stream S384): Cleavage Product Effluent from D-2304 Nozzle N4 via Level Control Valve LV-1
- Line 4"-P-2315-316L (Stream S387): Emergency Cumene Quench Header via SIL-2 Blowdown/Quench Valve XV-1309 to D-

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- Level Loop LIC-1301: Guided-Wave Radar LT-1301 (0-4500 mm, Normal 55%, LSHH 82%, LSLI 20%) modulates LV-1301 on
- Temperature Loop TIC-1302: Modulates Cooling Water Valve TV-1302 on E-2307 Shell Outlet (Normal 82.0 degC)
- SIS High-High Temperature Trip TXSHH-1301 (2oo3 voting, Setpoint 115.0 degC): Triggers Interlock I-2304 -> clos
- Pressure Loop PIC-1301: Pressure Transmitter PT-1301 (Normal 1.8 kg/cm2g, PAHH 5.5 kg/cm2g -> Interlock I-2304)
- Overpressure Protection: Dual Safety Relief Valves PSV-2304A / PSV-2304B (Set 11.0 kg/cm2g, Orifice 4J6) discha